SAHAYAK Presentation Guide

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Role: System Design, Technology and Workflow

Suggested speaking time: 4 to 5 minutes

Your job

You are the technical speaker. Explain what happens from the camera to the result on the screen. Show how the website, hand-point detector, backend and trained model work together. You do not need to read code line by line.

Slide 6 - The full workflow

Say:

Thank you, Barakha. I will explain what happens behind the screen. First, the user opens the website and allows camera access. MediaPipe Hands finds key points on the visible hands. The website changes the points into numbers and sends those numbers to the Python Flask backend. The trained model predicts a label, and the website shows the result.

The browser handles the camera preview and drawing of hand points. The backend runs the model prediction. These are separate jobs.

Slide 7 - What are hand landmarks?

Say:

A landmark is a key point on the hand, such as a fingertip or a joint. MediaPipe Hands can find 21 landmarks on each hand. Every point has x, y and z values. Those values describe where the hand points are, so a model can process them as numbers.

Think of it this way: a person sees a bent finger. The program describes the hand using points and numbers. The model learns patterns from labelled examples. It does not understand sign language in the same way a person does.

Slide 8 - Why 126 numbers for a static sign?

Say:

Each hand has 21 landmarks. Each landmark has three coordinates: x, y and z. So one hand gives 21 x 3 = 63 values. Two hands give 63 x 2 = 126 values.

Write on the board:

2 hands x 21 landmarks x 3 coordinates = 126 values.

These 126 values are the input size for the two-hand static model. They are not 126 signs and they do not tell us the model's accuracy.

Slide 9 - Why a sequence model?

Say:

Some signs include movement, so one frame may not show the full sign. SAHAYAK also has a model for continuous signs. Each frame has 132 values: 126 hand-point values and six wrist-position values. The system arranges the sequence into 10 time steps. 132 x 10 = 1,320 values.

This is the input format. It does not prove every moving sign works well. The model needs good labelled movement examples and testing.

Slide 10 - Frontend, backend and model

Say:

The frontend is the website shown in the browser and is hosted on Vercel. The backend is the Python Flask server hosted on Render. The website sends the prepared numbers to an API route. The backend loads a saved model and sends a prediction back to the website.

The current model files use scikit-learn-compatible models saved with Joblib. We should not call the system a TensorFlow or CNN system unless we have changed and checked the actual code and model.

Some translation or speech services need internet, so we should not say the whole project works fully offline.

Slide 11 - Data, training and fair testing

Say:

To add a new sign, we need correctly labelled examples. We check those examples, train a trial model, and test it on fresh examples that were not used for training. We should also check that the old signs still work. When possible, test using different people or different recording sessions.

A response can be fast but wrong. We should only give an accuracy percentage when we have a saved test report. Useful results include how often each sign was correct and which signs were confused.

Simple technical notes

- Static model: 126 values for a two-hand pose.
- Continuous model: 132 values per frame and 10 time steps, making 1,320 values.
- Frontend: HTML, CSS and JavaScript in the browser.
- Hand points: MediaPipe Hands in the browser.
- Backend: Python with Flask.
- Model: scikit-learn-compatible classifier saved with Joblib.
- Hosting: Vercel for the frontend, Render for the backend.

Questions judges may ask

Q: Why use landmarks instead of a full image?

A: Landmarks give a smaller set of numbers that describe hand shape and position. They can still miss other information that matters to sign language.

Q: Does adding a label automatically teach the model?

A: No. We need labelled examples, training, fair testing and deployment of an updated model.

Q: Where does recognition happen?

A: The browser finds and draws hand points. The trained model makes the prediction in the backend.

Q: What is confidence?

A: A score showing how strongly the model favours one label. It is not a guarantee that the result is correct.

Q: How do we know the model is accurate?

A: We need a proper test using fresh examples and results for each label. We should not guess a percentage.

Handover to Himanxhu

Say:

That is the path from the camera to the model prediction. My teammate Himanxhu V Ghimire will now demonstrate the website, explain how to test it and discuss what we still need to improve.

Speaker checklist

- Remember the workflow in order.
- Do not mix up the static model and the sequence model.
- Explain what 126, 132 and 1,320 mean.
- Use the active backend label list, not an old list from memory.
- Do not claim unmeasured accuracy.
- If a live request fails, explain the intended workflow honestly instead of pretending it worked.